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ELECTRONIC DEVICE USING PERSONAL AI MODEL, AND OPERATION METHOD THEREOF

Abstract

An electronic device includes a memory storing one or more instructions; a communication circuit; and a processor operatively coupled to the memory and the communication circuit, in which the one or more instructions, when executed by the processor, cause the electronic device to: identify a use pattern of specified media items from a plurality of media items stored in the memory, determine, based on the use pattern, a score of each of the specified media items, extract, using a main AI model stored in the memory, a feature corresponding to a characteristic of each of the specified media items, acquire a first AI model trained based on the score and the feature, based on the first AI model, determine a first preference of each of first media items from the plurality of media items, and based on the first preference, perform a function related to the first media items.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of PCT International Application No. PCT/KR2024/001344, which was filed on Jan. 29, 2024, and claims priority to Korean Patent Application No. 10-2023-0012484, filed on Jan. 31, 2023, and Korean Patent Application No. 10-2023-0020740, filed on Feb. 16, 2023, in the Korean Intellectual Property Office, the disclosures of each of which are incorporated by reference herein their entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to an electronic device using a personal AI model and an operation method thereof.

2. Description to Related Art

[0003] Currently, portable digital communication devices have become essential items. Consumers use portable digital communication devices to acquire various high-quality services at anytime and at any location.

[0004] A recognition service using an artificial intelligence (AI) technology may be a service of recognizing, based on an interface implemented by portable digital communication devices, various types of media by an AI model trained through a deep learning scheme, and providing consumers with various content services by using a result of the recognition. To provide a high-quality AI recognition service to consumers, a technology of accurately identifying the preference or interest of the user and a technology of providing an appropriate content service corresponding to the identified user's intention is required.

SUMMARY

[0005] According to one or more embodiments, an electronic device including: memory storing one or more instructions; a communication circuit; and a processor operatively coupled to the memory and the communication circuit, wherein the one or more instructions, when executed by the processor, cause the electronic device to: identify a use pattern of specified media items from a plurality of media items stored in the memory, determine, based on the use pattern, a score of each of the specified media items, extract, using a main AI model stored in the memory, a feature corresponding to a characteristic of each of the specified media items, acquire a first AI model trained based on the score and the feature, based on the first AI model, determine a first preference of each of first media items from the plurality of media items, and based on the first preference, perform a function related to the first media items.

[0006] According to one or more embodiments, an operation method of an electronic device including identifying a use pattern of specified media items from a plurality of media items stored in the electronic device; determining, based on the use pattern, a score of each of the specified media items; extracting, using a main AI model stored in the memory, a feature corresponding to a characteristic of each of the specified media items; acquiring a first AI model trained based on the

score and the feature; based on the first AI model, determining a first preference of each of first media items from the plurality of media items; and based on the first preference, performing a function related to the first media items.

[0007] According to one or more embodiments, a non-transitory computer readable medium having instructions stored therein, which when executed by a processor in an electronic device, cause the processor to execute a method including: identifying a use pattern of specified media items from a plurality of media items stored in the electronic device; determining, based on the use pattern, a score of each of the specified media items; extracting, using a main AI model stored in the memory, a feature corresponding to a characteristic of each of the specified media items; acquiring a first AI model trained based on the score and the feature; based on the first AI model, determining a first preference of each of first media items from the plurality of media items; and based on the first preference, performing a function related to the first media items.

[0008] DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a block diagram illustrating an electronic device in a network environment according to various embodiments.

[0010] FIG. 2A is a block diagram illustrating a system including an electronic device, multiple external electronic devices, and a server according to one or more embodiments.

[0011] FIG. 2B is a flowchart illustrating an operation method of an electronic device according to one or more embodiments.

[0012] FIG. 3 is a block diagram illustrating an AI service manager executed in an electronic device according to one or more embodiments.

[0013] FIG. 4A is a flowchart illustrating a method of training a personal AI model by an electronic device according to one or more embodiments.

[0014] FIG. 4B is a flowchart illustrating a method in which the electronic device trains a personal AI model by using information on media received from the outside according to one or more embodiments.

[0015] FIG. 5 illustrates a method of training a personal AI model by an electronic device according to one or more embodiments.

[0016] FIG. 6 illustrates a method in which an electronic device obtains a feature and a score for media for training a personal AI model according to one or more embodiments.

[0017] FIG. 7 is a flowchart illustrating a method of training a personal AI model by an electronic device when the state of the electronic device satisfies a specified condition according to one or more embodiments.

[0018] FIG. 8A is a flowchart illustrating a method in which an electronic device performs a function related to media items by using a personal AI model according to one or more embodiments.

[0019] FIG. 8B is a flowchart illustrating a method of classify media items by using a personal AI model by an electronic device according to one or more embodiments.

[0020] FIG. 9A illustrates an operation of identifying preferences of media items by using a personal AI model by an electronic device according to one or more embodiments.

[0021] FIG. 9B illustrates an operation of identifying preferences of media items by using an AI model of other person by an electronic device according to one or more embodiments.

[0022] FIG. 9C illustrates an operation of identifying preferences of media items by using an AI model for each group by an electronic device according to one or more embodiments.

[0023] FIG. 10 illustrates an AI service manager for managing multiple AI models according to one or more embodiments.

[0024] FIG. 11A is a flowchart illustrating a method of performing a function related to media items by using an AI model of other person according to one or more embodiments.

[0025] FIG. 11B is a flowchart illustrating a method of performing a function related to media items by using an AI model of a specific group according to one or more embodiments.

[0026] FIG. 12A is a flowchart illustrating a method of performing a function related to media items by using an AI model of a specific person according to one or more embodiments.

[0027] FIG. 12B is a flowchart illustrating a method of providing a guide to media by using an AI model of a specific person according to one or more embodiments.

[0028] FIG. 13A illustrates a method of identifying an image preferred by a user among multiple images by using a personal AI model according to one or more embodiments.

[0029] FIG. 13B illustrates a method of identifying an image preferred by other person among multiple images by using an AI model of other person according to one or more embodiments.

[0030] FIG. 13C illustrates a method of classifying images according to attributes of multiple images by using a personal AI model according to one or more embodiments.

[0031] FIG. 14 illustrates a method of sharing, using a group-specific AI model, images for each corresponding group according to one or more embodiments.

[0032] FIG. 15A illustrates a method of identifying an image preferred by an expert among multiple images by using an expert AI model according to one or more embodiments.

[0033] FIG. 15B illustrates a method of providing a guide to an image by using an expert AI model according to one or more embodiments.

Description

DETAILED DESCRIPTION

[0034] FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to one or more embodiments, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to one or more embodiments, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0035] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to one or more embodiments, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be

adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0036] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to one or more embodiments, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to one or more embodiments, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or In one or more examples, include a software structure other than the hardware structure.

[0037] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0038] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0039] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0040] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to one or more embodiments, the receiver may be implemented as separate from, or as part of the speaker.

[0041] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to one or more embodiments, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0042] The audio module **170** may convert a sound into an electrical signal and vice versa. According to one or more embodiments, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0043] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to one or more embodiments, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0044] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to one or more embodiments, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0045] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to one or more embodiments, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0046] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to one or more embodiments, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0047] The camera module **180** may capture a still image or moving images. According to one or more embodiments, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0048] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0049] The battery **189** may supply power to at least one component of the electronic device **101**. According to one or more embodiments, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0050] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to one or more embodiments, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the

first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0051] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to one or more embodiments, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0052] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to one or more embodiments, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to one or more embodiments, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to one or more embodiments, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0053] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to one or more embodiments, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0054] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0055] According to one or more embodiments, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to one or more embodiments, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the

function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to one or more embodiments, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0056] FIG. 2A is a block diagram illustrating a system including an electronic device, multiple external electronic devices, and a server according to one or more embodiments.

[0057] Referring to FIG. 2A, a system **200** according to one or more embodiments may include an electronic device **201**, multiple external electronic devices **202** and **203**, and a server **208**. For example, the electronic device **201** may be implemented to be identical or similar to the electronic device **101** of FIG. 1. The multiple external electronic devices **202** and **203** may be implemented to be identical or similar to the electronic devices **102** and **104** of FIG. 1. In addition, the server **208** may be implemented to be identical or similar to the server **108** of FIG. 1.

[0058] The electronic device **201** according to one or more embodiments may transmit or receive data to or from the multiple external electronic devices **202** and **203** and the server **208**. For example, the electronic device **201** may receive data of media and/or an AI model from the multiple external electronic devices **202** and **203**. In addition, the electronic device **201** may receive data of media and/or an AI model from the server **208**.

[0059] According to one or more embodiments, the electronic device **201** may include a processor **220**, memory **230**, a battery **240**, a communication module **250**, and a display **260**.

[0060] According to one or more embodiments, the processor **220** may control the overall operation of the electronic device **201**. The processor **220** may be implemented to be identical or similar to the processor **120** of FIG. 1. According to the implementation, the processor **220** may include a main processor (e.g., the main processor **121** of FIG. 1 or an application processor) or an auxiliary processor (e.g., the auxiliary processor **123** of FIG. 1 or a neural processing unit (NPU)) which can be operated independently or together with the main processor.

[0061] According to one or more embodiments, the processor **220** may execute an AI service manager (or an AI service manager module) for managing a service or a function utilizing artificial intelligence (AI) or an AI model. For example, the AI service manager may be executed in the foreground or the background. In one or more examples, the AI service manager may be a native application of the electronic device **201**. In one or more examples, the AI service manager may be a third party application that is downloaded to the electronic device **201**.

[0062] According to one or more embodiments, the memory **230** (e.g., the memory **130** of FIG. 1) may store data related to the media. For example, the electronic device **201** may refer to data related to an image, a video, audio which can be executed in the electronic device **201**. In addition, the memory **230** may store an application (e.g., a photo viewing application, a video reproduction application, an audio reproduction application, a file explorer application, a background setting application, and a file sharing application) related to the media.

[0063] According to one or more embodiments, the processor **220** may identify a use pattern of multiple media items among media items stored in the memory **230** (e.g., all media items stored in the memory **230**). For example, the processor **220** may identify or monitor an interaction with the electronic device **201** that a user performs such as viewing, sharing, editing, favorites setting,

background setting, and/or deleting of the media through the application related to the media. In one or more examples, external usage history data may be accessed by the AI service manager such as programs and content a user watches on TV. The external usage history may be downloaded and stored in the memory **230**. In one or more examples, the external usage history may be acquired via a near field communication link (e.g., Bluetooth) each time the electronic device **201** is paired with an external device. In one or more examples, the AI service manager may be configured to acquire the external usage history data via a Wi-Fi connection. The processor **220** may determine or identify, based on the user pattern of the media, a score of the corresponding media. For example, the score may refer to a score or a value based on the use pattern of the media. For example, the processor **220** may convert the use pattern of the media into the score by using a specified equation. For example, the processor **220** may determine that the higher the score of the media, the user uses the corresponding media more frequently or has more interest.

[0064] According to one or more embodiments, the memory **230** may store a main AI model. For example, the main AI model may be an AI model pre-stored in the electronic device **201**. For example, the main AI model may be an AI model which is a reference to an AI model (e.g., a personal AI model) personalized, based on the use pattern of the user, to the user of the electronic device **201**. For example, the processor **220** may train media determined that the user uses more frequently or has more interest with reference to the main AI model, and obtain (or generate or update) the personal AI model. For example, the processor **220** may store the trained personal AI model in the memory **230**. For example, the main AI model and the personal AI model may be implemented as a deep learning-type (e.g., a convolution neural network (CNN)-type) AI model. For example, the processor **220** may perform on-device AI model training and utilization. In one or more examples, the AI model may be trained remotely and downloaded to the electronic device **201**. In one or more examples, the AI model may be updated on the electronic device **201** based on usage data of the electronic device **201**. In one or more examples, the electronic device **201** may be updated with an updated AI model.

[0065] According to one or more embodiments, the processor **220** may extract or identify a feature of each of the multiple media items by using the main AI model stored in the memory **230**. For example, when media is input, the main AI model may output a feature corresponding to the media. The feature may be data representing a characteristic of the media recognized by the main AI model. For example, the feature may be represented as a sequence of floats (e.g., a 4 byte float*512), and may have a value varying according to the attribute (e.g., color, tone, color temperature, or composition) used to classify the media. However, the feature may have a value which does not enable backtracking of the characteristic of the media even when the feature is analyzed.

[0066] According to one or more embodiments, the processor **220** may train, based on the score and the feature of each of the multiple media items, the AI model. The processor **220** may obtain (or generate or update) a personal AI model personalized to the user of the electronic device **201** as a result of the training. For example, the processor **220** may train the personal AI model in the state in which the electronic device **201** is idle or the battery **240** (e.g., the battery **189** of FIG. 1) is being charged. In one or more examples, the media items may be content (e.g., TV shows, movies, Youtube videos, etc.) the user has watched or viewed (e.g., websites, articles, etc.).

[0067] According to one or more embodiments, the processor **220** may identify or determine a first preference of each of first media stored in the memory **230** by using the personal AI model. For example, the first preference may refer to a preference score or a preference value of each of the first media determined by the personal AI model personalized to the user of the electronic device **201**. For example, the first media may be all media items stored in the memory **230** or media items for which the user has requested identification among all media items. In one or more examples, the first media may be media satisfying a pre-configured criterion (e.g., media generated at a specific time point, generated for a specific time interval, generated at a specific place, or obtained

from a specific person or a specific device) among all media items stored in the memory **230**. The processor **220** may store the first preference of each of the first media in the memory **230**.

[0068] According to one or more embodiments, the processor **220** may perform, based on the first preference, a function related to the first media. For example, the processor **220** may recommend, based on the first preference, at least one media item among the first media items. For example, the processor **220** may recommend media having a higher preference to the user. For example, the processor **220** may also classify, based on the first preference, the first media. For example, the processor **220** may classify the first media in the order of the first preference.

[0069] According to one or more embodiments, the processor **220** may also classify the first media by using the personal AI model according to a specific person, a specific character, a specific place, or a specific time interval. For example, the processor **220** may separately classify media including a specific person (or a specific character), may separately classify media obtained at a specific place, or may separately classify media obtained during specific time interval.

[0070] According to one or more embodiments, the processor **220** may identify the attribute (e.g., color, tone, color temperature, or composition) of each of the first media by using the personal AI model. For example, the processor **220** may classify, based on the identified attribute, the first media. For example, the processor **220** may classify the first media into at least one media item based on the composition and at least one media item based on the tone or color temperature.

[0071] According to one or more embodiments, the processor **220** may receive, from at least one of the multiple external electronic devices **202** and **203** through the communication module **250**, information on the personalized AI model (e.g., an AI model of other person) of the user of the corresponding external electronic device. The processor **220** may store the AI model of other person in the memory **230**. The processor **220** may identify the preference of the media stored in the memory **230** by using the AI model of other person. The processor **220** may perform, based on the identified preference, a function related to the corresponding media. For example, the processor **220** may transmit the media preferred by other person among the media stored in the memory **230** to the external electronic device.

[0072] According to one or more embodiments, the electronic device **201** may receive information on a specific personal AI model of a specific person (e.g., expert or celebrity) from the server **208** through the communication module **250**. The processor **220** may store the specific AI model in the memory **230**. The processor **220** may identify the preference to the media stored in the memory **230** by using the specific AI model. The processor **220** may perform, based on the identified preference, a function related to the corresponding media. For example, the processor **220** may identify media preferred by the specific person among media stored in the memory **230**. In one or more examples, the processor **220** may compare the preference of the specific person with the preference of the user so as to provide a guide to the media through the display **260** (e.g., the display module **160** of FIG. 1).

[0073] As described above, the electronic device **201** may use the personalized personal AI model to recommend the media that matches the preference or interest or classify the media that matches the user's preference among all media items stored in the memory **230**. In addition, the electronic device **201** may easily identify the media that matches the preference of other person or matches the preference of an expert among all media items stored in the memory **230**.

[0074] FIG. 2B is a flowchart illustrating an operation method of an electronic device according to one or more embodiments.

[0075] Referring to FIG. 2B, at least a part of an operation performed by an electronic device (e.g., the electronic device **201** of FIG. 2A) may be performed by a processor (e.g., the processor **220** of FIG. 2A). However, for convenience of description, it will be described that the electronic device **201** performs the corresponding operation.

[0076] According to one or more embodiments, in operation **211**, the electronic device **201** may identify a use pattern of specified media items among all media items stored in memory (e.g., the

memory **230** of FIG. 2A), and determine, based on the identified use pattern, a score of each of the specified media items. For example, the specified media items may be selected by the user, or may be automatically selected by a processor (e.g., the processor **220** of FIG. 2A). For example, the specified media items may be at least a part of all media items. For example, the electronic device **201** may identify a use pattern in consideration of at least one of viewing, sharing, editing, deleting, favorites setting, or background setting of the specified media items.

[0077] According to one or more embodiments, in operation **213**, the electronic device **201** may extract a feature corresponding to a characteristic of each of the specified media items by using a main AI model stored in the memory **230**.

[0078] According to one or more embodiments, in operation **215**, the electronic device **201** may obtain a personalized first AI model trained based on the score and the feature. For example, the first AI model may be an AI model personalized by the user of the electronic device **201**.

[0079] According to one or more embodiments, in operation **217**, the electronic device **201** may determine, based on the first AI model, a first preference of each of multiple first media items. For example, the multiple first media items may be specified media items selected by the user directly, or automatically selected by the processor (e.g., the processor **220** of FIG. 2A) to determine the preference. For example, the multiple first media items may be at least a part of all media items stored in the memory **230**. In addition, the multiple first media items may include newly obtained media items (e.g., media items newly obtained from the outside or newly captured or filmed by a camera). According to the implementation, operations **215** and **217** may be temporally separated. For example, after the first AI model is obtained by operation **215**, operation **217** may not be performed immediately.

[0080] According to one or more embodiments, in operation **219**, the electronic device **201** may perform, based on the first preference, a function related to the multiple first media items. For example, the electronic device **201** may recommend at least a part of the multiple first media items. In one or more examples, the electronic device **201** may classify the multiple first media items by a category based on the first preference.

[0081] FIG. 3 is a block diagram illustrating an AI service manager executed in an electronic device according to one or more embodiments.

[0082] Referring to FIG. 3, according to one or more embodiments, the processor **220** (e.g., the processor **220** of FIG. 2A) may execute an AI service manager **301**. For example, the AI service manager **301** may be a module managing a service (or a function) using an AI model. For example, the AI service manager **301** may be executed in the foreground or the background of an electronic device **201** (e.g., the electronic device **201** of FIG. 2A).

[0083] According to one or more embodiments, the AI service manager **301** may include an event handler **310**, a logger **320**, and an AI framework **340**.

[0084] According to one or more embodiments, the event handler **310** may identify or monitor an event generated in the electronic device **201**. The AI service manager **301** may identify an event generated in the electronic device **201** through the event handler **310**, and may perform a specified operation based on the identified event. For example, the AI service manager **301** may perform an operation of training a personal AI model when the electronic device **201** is identified to be in an idle state. When it is identified that the electronic device **201** enters a specific place or generates a specific group, the AI service manager **301** may perform an operation of obtaining an AI model of the group. In addition, the AI service manager **301** may perform an operation of obtaining an expert AI model when the electronic device **201** performs a specific service or a specific function. Detailed methods of performing the corresponding operations by the electronic device **201** (or the AI service manager **301**) will be described below.

[0085] According to one or more embodiments, the event handler **310** may include a device event handler **311** and an action event handler **312**. For example, the device event handler **311** may obtain the state of the electronic device **201** and a new media item, and/or may identify or monitor an

event related to a function related to media. The action event handler **312** may identify or monitor an event in which the electronic device **201** enters a specific place or generates a group with at least one external electronic device.

[0086] According to one or more embodiments, the logger **320** may identify a use pattern of a specific media item **305**. For example, the use pattern of the specific media item may be based on an interaction (e.g., viewing, sharing, background setting, favorites setting, indexing, editing, and deleting) for the specific media item **305**. For example, the logger **320** may identify a viewing time and/or a viewing count of the specific media item **305**. The logger **320** may identify a transmission counter, an upload place, and/or a copy location of the specific media item **305**. The logger **320** may identify whether a tag is specified, whether the background setting is made, or the favorites setting is made for the specific media item **305**. The logger **320** may identify an edition count and/or an edition level of the specific media item **305**. The logger **320** may identify whether the specific media item **305** is deleted and/or hidden. In one or more example, the content of a media item may be identified based on metadata associated with the media item. For example, the metadata may indicated a genre or category associated with the media item (e.g., sports, music, news, etc.). In one or more examples, content analysis may be performed to determine the content of the media item. For example, an image recognition process may be performed on an image associated with the media content to determine the genre or category associated with the media item.

[0087] According to one or more embodiments, the logger **320** may identify a score based on the use pattern of the specific media item **305**. For example, the score may indicate the degree of interest (or an interest value) of the user for the specific media item **305**. The logger **320** may provide the score for the specific media item **305** to an AI framework **340**.

[0088] According to one or more embodiments, the AI framework **340** may execute and manage a main AI model **350** and a personal AI model **360**.

[0089] According to one or more embodiments, the main AI model **350** may extract a feature for the specific media item **305**. According to the implementation, the main AI model **350** may extract (or determine) an aesthetic score for the specific media item **305**. For example, the aesthetic score may be a score or a value indicating the characteristic (e.g., the view angle, composition, color, or color temperature) of the specific media item **305**.

[0090] According to one or more embodiments, the AI framework **340** may train the personal AI model by using the score and the feature for the specific media item **305**. For example, the AI framework **340** may train the personal AI model so that a score for the specific media item **305** is output when the feature for the specific media item **305** is input, with reference to the main AI model. According to the implementation, the AI framework **340** may train the personal AI model by further using the aesthetic score. For example, the AI framework **340** may train the personal AI model **360** so that the score for the specific media item **305** is output when the feature and the aesthetic score for the specific media item **305** is input, with reference to the main AI model **350**.

[0091] According to one or more embodiments, the AI framework **340** may obtain and store the personal AI model **360** as a result of the training. The AI framework **340** may store information on the personal AI model **360** in a personal data (DB) **330**. For example, the personal DB may store data (e.g., the use pattern, score, feature, and aesthetic score of the specific media item **350**) used to train the personal AI model. For example, the personal database **330** may be implemented as memory (e.g., the memory **230** of FIG. 2A).

[0092] According to the above-described method, the AI framework **340** may obtain the score, feature, and/or aesthetic score of each of the multiple media items stored in the memory **230**. The AI framework **340** may train, based on the obtained score, feature, and/or aesthetic score, the personal AI model.

[0093] According to one or more embodiments, the AI framework **340** may extract or identify a preference score of the media by using the personal AI model **360**. For example, the preference

score may be a value indicating the degree of the user's preference determined by the personal AI model **360**. The AI framework **340** may extract or identify the preference score of each of all media items stored in the memory **230** by using the personal AI model **360**. In one or more examples, the personal AI model **360** may extract or identify, using the personal AI model **360**, the preference score of the media for which the user has requested identification.

[0094] The conventional electronic device was limited to recommending the same media to everyone even though each user has different preferences. For example, when a title of a specific movie is input, the electronic device was able to recommend movies having similar titles or having ratings similar to the corresponding movie even though a specific individual likes or dislikes the specific movie. To overcome these disadvantages, the embodiments of the present disclosure are directed to using one or more AI models that are trained to a user's preferences.

[0095] An electronic device using the conventional AI model was able to inquire of the individual about preference of the specific media item or a score indicating the preference. However, in this case, the electronic device needed an additional UI/UX for receiving information from the user. In addition, the electronic device needed more logic to select appropriate media, the preference of which is to be asked to the user. In one or more examples, the electronic device was able to obtain personal information such as a user profile from the user. In this case, a security risk for the user personal information may occur.

[0096] The electronic device **201** according to one or more embodiments may utilize media stored in the electronic device **201** to train an AI model to which the user's preference is reflected, and may provide various functions (e.g., a media curation service, a media sharing service, a media recommendation service, and a media classification service) related to the media through the trained AI model.

[0097] In addition, the electronic device **201** according to one or more embodiments may utilize, as data for AI model training, media preferred by the user or media having high interest of the user to recommend content among various media stored in the electronic device **201**. To this end, the electronic device **201** may effectively select media for AI model training.

[0098] In addition, the electronic device **201** according to one or more embodiments may share the personalized AI model with other electronic devices. In this case, the electronic device **201** may share the user's preference through sharing of the personalized AI model even when personal information such as a user profile is not shared.

[0099] At least a part of the operations of the electronic device **201** described below may be performed by the processor **220**. However, for convenience of description, it will be described that the operations are performed by the electronic device **201**.

[0100] FIG. **4A** is a flowchart illustrating a method of training a personal AI model by an electronic device according to one or more embodiments.

[0101] Referring to FIG. **4A**, according to one or more embodiments, in operation **401**, an electronic device (e.g., the electronic device **201** of FIG. **2A**) may identify a request for generation of a personal AI model. For example, the electronic device **201** may identify, based on a specified event (e.g., a preference identification request or media classification request event of media) or a user input (e.g., an input of requesting generation of a personal AI model), a request for generation of a personal AI model.

[0102] According to one or more embodiments, in operation **403**, the electronic device **201** may determine a media set for training of the personal AI model among multiple media items (e.g., all media items) stored in memory (e.g., the memory **230** of FIG. **2A**). For example, the media set may include specified media items among all media items. For example, the specified media items may include a specified number of media. For example, the specified media items may be determined as media items having high frequency of use among all media items.

[0103] According to one or more embodiments, in operation **405**, the electronic device **201** may obtain (or identify), through a logger (e.g., the logger **320** of FIG. **3**), a score for a use pattern of

each of the media included in the media set.

[0104] According to one or more embodiments, in operation **407**, the electronic device **201** may obtain, through a main AI model (e.g., the main AI model **350** of FIG. **3**), a feature of each of the media included in the media set. According to the implementation, the electronic device **201** may further obtain an aesthetic score of each of the media included in the media set.

[0105] According to one or more embodiments, in operation **409**, the electronic device **201** may train, based on the score and the feature, the personal AI model. According to the implementation, the electronic device **201** may train the personal AI model by using the aesthetic score.

[0106] According to one or more embodiments, in operation **411**, the electronic device **201** may obtain the personal AI model as a result of the training, and store the obtained personal AI model.

[0107] FIG. **4B** is a flowchart illustrating a method in which the electronic device trains a personal AI model by using information on media received from the outside according to one or more embodiments.

[0108] Referring to FIG. **4B**, according to one or more embodiments, in operation **431**, an electronic device (e.g., the electronic device **201** of FIG. **2A**) may obtain information on at least one media item from an external electronic device (e.g., the external electronic device **202** or **203** of FIG. **2A**). In addition, the electronic device **201** may obtain information on at least one media item from a server (e.g., the server **208** of FIG. **2A**). For example, the information on at least one media item may include information on at least one media item or a feature of the at least one media item.

[0109] According to one or more embodiments, in operation **433**, the electronic device **201** may train, based on the obtained information, the personal AI model. For example, when the personal AI model is trained using a feature of a specific character (or a specific object), the personal AI model may more accurately recognize the media including the specific character. According to the implementation, the electronic device **201** may train the personal AI model by using the feature of the obtained media even though the electronic device does not directly obtain the media from the external electronic device **202** or **203** or the server **208**.

[0110] Accordingly, the electronic device **201** may reinforce the performance (e.g., recognition on a specific character or a specific object) of the personal AI model.

[0111] According to one or more embodiments, the electronic device **201** may obtain an AI model of other person, trained for a specific character or a specific object, from the external electronic device **202** or **203** or the server **208**. The electronic device **201** may more accurately recognize the media including the specific character by using the AI model of other person.

[0112] FIG. **5** illustrates a method of training a personal AI model by an electronic device according to one or more embodiments.

[0113] Referring to part (a) of FIG. **5**, according to one or more embodiments, a main AI model (e.g., the main AI model **350** of FIG. **3**) may extract or output a feature **306** corresponding to a specific media item **305** when the specific media item **305** is input. According to the implementation, when the specific media item **305** is input, the electronic device **201** may extract or output the feature **306** and an aesthetic score **307** corresponding to the specific media item **305**.

[0114] Referring to part (b) of FIG. **5**, according to one or more embodiments, when the specific media item **305** is input, a logger (e.g., the logger **320** of FIG. **3**) may extract or output a score **308** based on a use pattern of the specific media item **305**.

[0115] Referring to part (c) of FIG. **5**, according to one or more embodiments, an operation of training a personal AI model (e.g., the personal AI model **360** of FIG. **3**) by using the feature **306** and the score **307** may be performed. According to the implementation, an operation of training the personal AI model **360** by using the aesthetic score **307** may be performed.

[0116] FIG. **6** illustrates a method in which an electronic device obtains a feature and a score for media for training a personal AI model according to one or more embodiments.

[0117] Referring to FIG. **6**, according to one or more embodiments, an electronic device (e.g., the

electronic device **201** of FIG. 2A or the logger **320** of FIG. 3) may identify a use pattern of an image **601**. The electronic device **201** may obtain, based on a use pattern of an image **601**, use pattern information **610**. For example, the use pattern information **610** may include at least a part of information on the number of times of viewing the image **601**, the number of times of sharing of the image, whether the image is set as background, whether the image is edited, or whether the image is deleted.

[0118] According to one or more embodiments, the electronic device **201** may normalize the use pattern information **610** to obtain normalized use pattern information **620**. For example, the electronic device **201** may normalize the user pattern information **610** according to a specified rule. For example, the normalized use pattern information **620** may include values for the use pattern of the media **601**.

[0119] According to one or more embodiments, the electronic device **201** may convert values included in the normalized use pattern information **620** into a score **630** by using a specified equation. For example, the score **630** may a score or a value based on the use pattern of the image **601**. For example, the specified equation may be determined by a sum (e.g., a total sum) of the values for the use pattern of the media **601**.

[0120] According to one or more embodiments, the electronic device **201** may obtain a feature **640** corresponding to the image **601** by using a main AI model **350**.

[0121] According to the above-described method, the electronic device **201** may obtain the score **630** and the feature **640** for training the personal AI model.

[0122] FIG. 7 is a flowchart illustrating a method of training a personal AI model by an electronic device when the state of the electronic device satisfies a specified condition according to one or more embodiments.

[0123] Referring to FIG. 7, according to one or more embodiments, in operation **701**, an electronic device (e.g., the electronic device **201** of FIG. 2A) may identify the state of the electronic device **201**.

[0124] According to one or more embodiments, in operation **703**, the electronic device **201** may identify whether the state of the electronic device **201** is an idle state. For example, the idle state may refer to the state in which the user is not disturbed while using the electronic device **201**. For example, the idle state may refer to the state in which the electronic device **201** is not being used or is being charged. The electronic device **201** may identify whether the electronic device **201** is in an idle state (e.g., the state in which there is no motion detection) through a sensor included in the electronic device **201**. In one or more examples, the electronic device **201** may identify, based on whether a battery (e.g., the battery **240** of FIG. 2A) is being charged, whether the electronic device **201** is in the idle state. In one or more examples, the idle state may be pre-determined as a particular time of day such as between midnight and 4:00 am. In one or more examples, the idle state may be pre-determined as a particular location (e.g., GPS location of electronic device **201**).

[0125] According to one or more embodiments, when the state of the electronic device **201** is identified to be the idle state (e.g., if yes in operation **703**), the electronic device **201** may identify, in operation **705**, whether a specified condition for updating a personal AI model is satisfied. For example, when a predetermined number of media items are newly obtained, a specified time has elapsed after the previous update, a specific period (e.g., a new media generation period and/or an idle state entering cycle) based on the user pattern of the user has elapsed, or when execution of a function related to the media is requested (e.g., identification of a media preference is requested or generation of a personal AI model for the media is requested), the electronic device **201** may determine that the specified condition is satisfied.

[0126] According to one or more embodiments, in operation **707**, the electronic device **201** may update the personal AI model. For example, the electronic device **201** may adjust the weight applied to the personal AI model to update the personal AI model. For example, the electronic device **201** may obtain a score based on a user pattern of each of newly obtained media items, and

obtain a feature of each of the newly obtained media items. Hereinafter, the electronic device **201** may additionally train the personal AI model by using the score and the feature of each of the newly obtained media items. The electronic device **201** may obtain weight values to be newly applied to the personal AI model as a result of the additional training.

[0127] According to one or more embodiments, the electronic device **201** may not update the personal AI model when both the conditions of operations **703** and **705** are not satisfied. However, according to the implementation, the electronic device **201** may update the personal AI model when any one of the conditions of operations **703** and **705** are satisfied.

[0128] According to the above-described method, the electronic device **201** may update the personal AI model so that the preference and the interest of the user are matched.

[0129] FIG. **8A** is a flowchart illustrating a method in which an electronic device performs a function related to media items by using a personal AI model according to one or more embodiments.

[0130] Referring to FIG. **8A**, according to one or more embodiments, in operation **801**, the electronic device **201** may detect a request for identification of a preference of a specific media item.

[0131] According to one or more embodiments, in operation **803**, the electronic device **201** may obtain a feature of each of media items by using a main AI model (e.g., the main AI model **350** of FIG. **3**). For example, the electronic device **201** may input media to the main AI model **350** to obtain the feature corresponding to the media.

[0132] According to one or more embodiments, in operation **805**, the electronic device **201** may obtain a preference of each of the media items by using a personal AI model (e.g., the personal AI model **360** of FIG. **3**). For example, the electronic device **201** may input a feature corresponding to the media to the personal AI model **360** to obtain a preference of the media.

[0133] According to one or more embodiments, in operation **807**, the electronic device **201** may perform, based on the preference obtained using the personal AI model **360**, a function related to media items. For example, the electronic device **201** may recommend, based on the preference, at least one of media items to the user (e.g., recommend at least one media item that matches the preference of the user). In one or more examples, the electronic device **201** may classify the media items for each category based on the preference. In one or more examples, the electronic device **201** may classify the media items for each category based on the attributes of the media items.

[0134] FIG. **8B** is a flowchart illustrating a method of classify media items by using a personal AI model by an electronic device according to one or more embodiments.

[0135] Referring to FIG. **8B**, according to one or more embodiments, in operation **831**, an electronic device (e.g., the electronic device **201** of FIG. **2A**) may select some media items according to a score based on a use pattern of each of all media items stored in the memory **230**.

[0136] According to one or more embodiments, in operation **833**, the electronic device **201** may identify the attribute of each of the selected media items. For example, the attribute may include the composition (e.g., horizontal, vertical, or diagonal), tone (e.g., blue, red, green, or white), and/or color temperature (e.g., hot, warm, mild, or cold) of the media (e.g., image). The electronic device **201** may determine a score or a value for the corresponding attributes for each of the selected media items.

[0137] According to one or more embodiments, in operation **835**, the electronic device **201** may classify, based on the identified attributes, the media items. For example, the electronic device **201** may classify media items each having a high score in the composition as composition-focused media items, and may classify media items each having a high score in the tone/color temperature as tone/color temperature-focused media items.

[0138] Through the above-described method, the electronic device **201** may provide a curation service for the media items.

[0139] FIG. **9A** illustrates an operation of identifying preferences of media items by using a

personal AI model by an electronic device according to one or more embodiments.

[0140] Referring to FIG. 9A, according to one or more embodiments, an AI framework **340** may extract or identify first preference scores **910** for multiple first media items **905**, respectively, by using a personal AI model **360**. The AI framework **340** may generate a personal AI model or use a personal AI model (e.g., the personal DB **330** of FIG. 3) stored in the memory **230**. For example, the first preference scores **910** may be scores indicating preferences of the corresponding media items determined by the personal AI model **360** personalized to a user of an electronic device (e.g., the electronic device **201** of FIG. 2A). For example, the multiple first media items **950** may be at least some of all media items stored in the memory **230**.

[0141] According to one or more embodiments, the electronic device **201** may select and recommend at least one media item among the multiple first media items **905** by using the preference scores identified using the personal AI model. Accordingly, the electronic device **201** may automatically recommend a media item that matches the preference of the user among the multiple first media items.

[0142] FIG. 9B illustrates an operation of identifying preferences of media items by using an AI model of other person by an electronic device according to one or more embodiments.

[0143] Referring to FIG. 9B, according to one or more embodiments, the AI framework **340** may use an AI model of other person, received or downloaded from an external electronic device (e.g., **202**, **203**, or **208** of FIG. 2A), rather than the personal AI model **360** of FIG. 9A. For example, the AI framework **340** may extract or identify second preference scores **920** for multiple first media items **905**, respectively, by using an AI model **370** of other person. For example, the second preference scores **920** may be scores indicating preferences of the corresponding media items determined by the AI model **370** of other person, personalized to a user (e.g., other person) of an electronic device (e.g., the external electronic device **202** or **203** of FIG. 2A). In one or more examples, the second preference scores **920** may be scores indicating the preferences of the corresponding media items determined by the AI model **307** of other person, personalized to an expert or a specific persona (e.g., celebrity) stored in a server (e.g., the server **208** of FIG. 2A).

[0144] According to one or more embodiments, the electronic device **201** may select and recommend at least one of multiple first media items **905** by using preference scores determined by the AI model of other person. For example, the electronic device **201** may provide information on the media that matches the preference of other person or the expert among the multiple first media items. Accordingly, the user can easily identify the media that matches the preference of other person or the expert through the electronic device **201**.

[0145] FIG. 9C illustrates an operation of identifying preferences of media items by using an AI model for each group by an electronic device according to one or more embodiments.

[0146] Referring to FIG. 9C, according to one or more embodiments, an AI framework **340** may use group-specific AI model. For example, the AI framework **340** may manage and/or execute AI models **381**, **382**, and **383** for multiple groups.

[0147] According to one or more embodiments, the AI framework **340** may extract or identify preferences of media items related to a corresponding group among multiple first media items **905** by using the group AI model **381**, **382**, or **383** for each of the multiple groups. For example, the first group AI model **381** may output a first group preference **921** for at least one media item A **911** related to a first group among the multiple first media items **905**. For example, the first group preference **921** may be a score indicating a preference of at least one media item A **911** determined by the first group AI model **381**. For example, the second group AI model **382** may output a second group preference **922** for at least one media item B **912** related to a second group among the multiple first media items **905**. For example, the second group preference **922** may be a score indicating the preference of the at least one media item B **912** determined by the second group AI model **382**. For example, the third group AI model **383** may output a third group preference **923** for at least one media item C **913** related to a third group among the multiple first media items **905**.

For example, the third group preference **923** may be a score indicating the preference of the at least one media item **C 913** determined by the third group AI model **383**.

[0148] According to one or more embodiments, the electronic device **201** may share at least one of the multiple first media items **905** with the corresponding group by using the preference score determined by the group-specific AI model **381**, **382**, or **383**. Accordingly, the electronic device **201** easily share media that matches the preference for each group with external electronic devices included in the corresponding group. In addition, the user can easily identify and manage media that matches the preference for each group through the electronic device **201**.

[0149] FIG. **10** illustrates an AI service manager for managing multiple AI models according to one or more embodiments.

[0150] Referring to FIG. **10**, according to one or more embodiments, an AI service manager **1060** (e.g., the AI service manager **301** of FIG. **3**) may manage multiple AI models **1061**, **1062**, **1063**, and **1064**. For example, the first AI model **1061** may be a personal AI model personalized by a user of the electronic device **201**. The second AI model **1062** may be an AI model of other person, personalized by a user (e.g., other person) of an external electronic device (e.g., the external electronic device **202** or **203** of FIG. **2A**). The expert AI model **1063** may be an expert AI model personalized by an expert, obtained from a server (e.g., the server **208** of FIG. **2A**). The group AI model **1064** may be a group AI model for a group (e.g., a family group, a friend group, or a specific group) including multiple external electronic devices forming a communication connection with the electronic device **201** through communication technology. For example, the group AI model **1064** may include a separate group AI model for each group generated by the electronic device **201**.

[0151] According to one or more embodiments, the AI service manager **1060** may manage training DBs **1071**, **1072**, **1073**, and **1074** having trained the multiple AI models **1061**, **1062**, **1063**, and **1064**. For example, the training DBs **1071**, **1072**, **1073**, and **1074** may store data (e.g., a feature, a score, or an aesthetic score of media) having trained a corresponding AI model.

[0152] FIG. **11A** is a flowchart illustrating a method of performing a function related to media items by using an AI model of other person according to one or more embodiments.

[0153] Referring to FIG. **11A**, according to one or more embodiments, in operation **1101**, an electronic device (e.g., the electronic device **201** of FIG. **2A**) may obtain an AI model of other person from an external electronic device (e.g., the external electronic device **202** or **203** of FIG. **2A**). For example, the AI model of other person may be an AI model personalized by a user (e.g., other person) of the external electronic device **202** or **203**.

[0154] According to one or more embodiments, in operation **1103**, the electronic device **201** may obtain, using a main AI model (e.g., the main AI model **350** of FIG. **3**), features of media items for which the preferences are to be identified. For example, the main AI model may receive media and output a feature of the media.

[0155] According to one or more embodiments, in operation **1105**, the electronic device **201** may identify, using an AI model of other person, preferences of media items for which the preferences are to be identified. For example, the AI model of other person may receive features of the obtained media items through the main AI model and output the preference of the corresponding media items. For example, the output preferences may be values each indicating the interest or preference of other person to the corresponding media.

[0156] According to one or more embodiments, in operation **1107**, the electronic device **201** may perform, based on the identified preferences, a function related to media items. For example, the electronic device **201** may transmit a specified number of media items having high preference among the media items stored in the memory **230** to the external electronic device **202** or **204**. In one or more examples, the electronic device **201** may classify, as a category, the specified number of media items having high preferences among the media items stored in the memory **230**.

[0157] According to one or more embodiments, the electronic device **201** may search for at least

one external electronic device to which AI model of the electronic device can be shared. In this case, when multiple external electronic devices are found, the electronic device **201** may provide a list of external electronic devices. For example, the list of external electronic devices may include a representative image of an AI model shared among the external electronic devices. The electronic device **201** may obtain, from the selected external electronic device, an AI model of other person, personalized by the corresponding external electronic device.

[0158] FIG. **11B** is a flowchart illustrating a method of performing a function related to media items by using an AI model of a specific group according to one or more embodiments.

[0159] Referring to FIG. **11B**, according to one or more embodiments, in operation **1131**, an electronic device (e.g., the electronic device **201** of FIG. **2A**) may form a group with external electronic devices (e.g., **202** and **203** of FIG. **2A**).

[0160] According to one or more embodiments, in operation **1133**, the electronic device **201** may share an AI model with external electronic devices included in the group. For example, the electronic device **201** may transmit a personal AI model to the external electronic devices. In addition, the electronic device **201** may also obtain an AI model from the external electronic devices. In one or more examples, the electronic device **201** may separately generate a group AI model for the group.

[0161] According to one or more embodiments, in operation **1135**, the electronic device **201** may identify preferences of media items related to a group by using a shared AI model. The electronic device **201** may share the identified preferences with external electronic devices included in the group.

[0162] According to one or more embodiments, in operation **1137**, the electronic device **201** may select, based on the identified preferences, at least one media item to be shared with external electronic devices from among media items related to the group.

[0163] According to one or more embodiments, in operation **1139**, the electronic device **201** may share at least one selected media item with external electronic devices included in the group. For example, the electronic device **201** may transmit at least one media item selected from the external electronic device included in the group.

[0164] FIG. **12A** is a flowchart illustrating a method of performing a function related to media items by using an AI model of a specific person according to one or more embodiments.

[0165] Referring to FIG. **12A**, according to one or more embodiments, in operation **1201**, an electronic device (e.g., the electronic device **201** of FIG. **2A**) may obtain an AI model of a specific person from a server (e.g., the server **208** of FIG. **2A**). For example, the specific person may include, but not limited to, an external or a celebrity. In this case, the AI model of the specific person may be an AI model personalized by an expert or a celebrity. The AI model of the specific person may be an AI model stored or registered in the server **208** in advance by the specific person.

[0166] According to one or more embodiments, in operation **1203**, the electronic device **201** may obtain features of media items by using a main AI model. For example, the main AI model may receive an input of media and output a feature of the corresponding media.

[0167] According to one or more embodiments, in operation **1205**, the electronic device **201** may identify preferences of media items by using the AI model of the specific person. For example, the AI model of the specific person may receive features for media items obtained through the main AI model, and output the preferences of the corresponding media items. For example, the output preferences may be a value indicating the interest or preference of the specific person to the corresponding media.

[0168] According to one or more embodiments, in operation **1207**, the electronic device **201** may perform, based on the identified preferences, a function related to media items. For example, the electronic device **201** may classify, as a separate category, a specified number of media items having high preferences among the media items stored in the memory **230**. In one or more examples, the electronic device **201** may recommend, as expert-preferred media or celebrity-

preferred media, at least one media item having a high preference among the media items stored in the memory **230**.

[0169] FIG. **12B** is a flowchart illustrating a method of providing a guide to media by using an AI model of a specific person according to one or more embodiments.

[0170] Referring to FIG. **12B**, according to one or more embodiments, in operation **1231**, an electronic device (e.g., the electronic device **201** of FIG. **2A**) may identify a first preference of a specific media item by using an AI model of a specific person. For example, the AI model of the specific person may be obtained from a server (e.g., the server **208** of FIG. **2A**). For example, the first preference may be a score or a value indicating the preference to the specific media item determined by the AI model of the specific person.

[0171] According to one or more embodiments, in operation **1233**, the electronic device **201** may identify a second preference of a specific media item by using a personal AI model. The second preference may be a score or a value indicating the preference to the specific media item determined by the personal AI model.

[0172] According to one or more embodiments, in operation **1235**, the electronic device **201** may compare the first preference and the second preference to provide a guide to the specific media item. For example, the electronic device **201** may identify a difference between the first preference and the second preference. The electronic device **201** may identify a difference between the first preference and the second preference with reference to the first preference, and may provide, based on the identified difference, a guide to the specific media item. For example, the guide may include whether a specific media item is preferred (e.g., information indicating whether media is preferred or unpreferred), the degree of preference (e.g., a preference score for media), and/or correction information (e.g., whether media needs to be corrected and information on a correction method of media). For example, when an AI model of a specific person is an AI model of a photo expert and media is an image, the electronic device **201** may provide a guide to the view angle, exposure, focus, and/or white balance of the corresponding image. For example, the electronic device **201** may provide, based on the difference between the first preference and the second preference, a correction method for at least one of the view angle, exposure, focus, and white balance of the corresponding media as a guide.

[0173] FIG. **13A** illustrates a method of identifying an image preferred by a user among multiple images by using a personal AI model according to one or more embodiments.

[0174] Referring to FIG. **13A**, according to one or more embodiments, an electronic device (e.g., the electronic device **201** of FIG. **2A**) may identify preferences of multiple images **1310**, **1320**, and **1330**, respectively, by using a personal AI model **1350** (e.g., the personal AI model **360** of FIG. **3**). The electronic device **201** may determine as a recommended image, image A **1310** having the highest preference among the multiple images **1310**, **1320**, and **1330**.

[0175] Accordingly, the electronic device **201** may effectively identify an image that matches the user's preference among the multiple images.

[0176] FIG. **13B** illustrates a method of identifying an image preferred by other person among multiple images by using an AI model of other person according to one or more embodiments.

[0177] Referring to FIG. **13B**, according to one or more embodiments, an electronic device (e.g., the electronic device **201** of FIG. **2A**) may identify preferences of multiple images **1310**, **1320**, and **1330**, respectively, by using an AI model **1355** of other person (e.g., the AI model **370** of other person of FIG. **9**). The electronic device **201** may determine, as a recommended image, image B **1320** having the highest preference among the multiple images **1310**, **1320**, and **1330**.

[0178] Accordingly, the electronic device **201** may effectively identify an image that matches the preference of other person among the multiple images. In addition, the electronic device **201** may identify information on the preference, interest, or taste of other person through the AI model of other person even though personal information of other person is not obtained.

[0179] FIG. **13C** illustrates a method of classifying images according to attributes of multiple

images by using a personal AI model according to one or more embodiments.

[0180] Referring to FIG. 13C, according to one or more embodiments, an electronic device (e.g., the electronic device **201** of FIG. 2A) may obtain the attribute (e.g., composition, tone, or color temperature) of each of multiple images **1310**, **1320**, and **1330** by using a personal AI model **1355** (e.g., the personal AI model **360** of FIG. 3). The electronic device **201** may determine a score for each attribute for each of the multiple images **1310**, **1320**, and **1330** by using the personal AI model **1355**. The electronic device **201** may classify the multiple images **1310**, **1320**, and **1330** according to the attribute. For example, the electronic device **201** may classify images **1310** and **1320** each having a high score in the composition among the multiple images **1310**, **1320**, and **1330** as composition-focused images. In addition, the electronic device **201** may classify images **1320** and **1330** each having a high score indicating the tone/color temperature among the multiple images **1310**, **1320**, and **1330** as tone/color temperature-focused images.

[0181] According to the above-described method, the electronic device **201** may classify the multiple images for each attribute. In addition, the electronic device **201** may also provide information on an attribute, by which the images are classified. Accordingly, the electronic device **201** may provide a curation service to the user.

[0182] FIG. 14 illustrates a method of sharing, using a group-specific AI model, images for each corresponding group according to one or more embodiments.

[0183] Referring to FIG. 14, according to one or more embodiments, an electronic device (e.g., the electronic device **201** of FIG. 2A) may identify preferences of multiple images **1410**, **1420**, **1430**, and **1440**, respectively, by using group AI models **1450** and **1460** (e.g., group AI models **381**, **382**, and **383** of FIG. 9C).

[0184] According to one or more embodiments, the electronic device **201** may identify preferences of images **1410**, **1420**, and **1430** related to a first group, respectively, among the multiple images **1410**, **1420**, **1430**, and **1440** by using a first group AI model **1450**. For example, the electronic device **201** may share, based on the identified preferences, image A **1410** and image C **1430** among the images **1410**, **1420**, and **1430** with external electronic devices included in the first group.

[0185] According to one or more embodiments, the electronic device **201** may identify preferences of images **1420**, **1430**, and **1440** related to a second group, respectively, among the multiple images **1410**, **1420**, **1430**, and **1440** by using a second group AI model **1460**. For example, the electronic device **201** may share, based on the identified preferences, image B **1420** and image D **1440** among the images **1420**, **1430**, and **1440** with external electronic devices included in the second group.

[0186] According to the above-described method, the electronic device **201** may share interested images according to a group among the multiple images with external electronic devices included in the corresponding group. In addition, the electronic device **201** may identify information on the preference, interest, or taste of the group through a group AI model even though personal information of the users included in the group is not obtained.

[0187] FIG. 15A illustrates a method of identifying an image preferred by an expert among multiple images by using an expert AI model according to one or more embodiments.

[0188] Referring to FIG. 15A, according to one or more embodiments, an electronic device (e.g., the electronic device **201** of FIG. 2A) may identify preferences of multiple images **1510**, **1520**, and **1530**, respectively, by using an expert AI model **1550** (e.g., the AI model **370** of other person of FIG. 9). The electronic device **201** may determine image C **1530** having the highest preference among the multiple images **1510**, **1520**, and **1530** as an expert recommend image.

[0189] Accordingly, the electronic device **201** may effectively identify an image that matches the preference of an expert among the multiple images. In addition, the electronic device **201** may identify information on the preference, interest, or taste of the expert through an expert AI model without directly meeting an expert or obtaining personal information of the expert.

[0190] FIG. 15B illustrates a method of providing a guide to an image by using an expert AI model according to one or more embodiments.

[0191] Referring to FIG. 15B, according to one or more embodiments, an electronic device (e.g., the electronic device **201** of FIG. 2A) may identify a first preference of a specific image **1540** by using an expert AI model **1550** (e.g., the AI model **370** of other person of FIG. 9). The electronic device **201** may identify a second preference of a specific image **1540** by using a personal AI model **1560**. The electronic device **201** may provide guide information of the specific image **1540** by comparing the first preference and the second preference. For example, the electronic device **201** may provide, based on a difference between the first preference and the second preference, guide information of the specific image **1540**. For example, the electronic device **201** may provide, based on the difference between the first preference and the second preference, an improvement method for at least one of the view angle, exposure, focus, and white balance of the specific image **1540**.

[0192] By utilizing the above-described method, the electronic device **201** may provide an education service or an entertainment service by using an expert AI model.

[0193] An electronic device **201** according to one or more embodiments may include memory **230**, a communication circuit **250**, a processor **220**. The processor according to one or more embodiments may be configured to identify a use pattern of specified media items among all media items stored in the memory, and determine, based on the use pattern, a score of each of the specified media items. The processor according to one or more embodiments may be configured to identify a feature corresponding to a characteristic of each of the specified media items by using a main AI model stored in the memory. The processor according to one or more embodiments may be configured to acquire a personalized first AI model trained based on the score and the feature. The processor according to one or more embodiments may be configured to determine, based on the first AI model, a first preference of each of multiple first media items. The processor according to one or more embodiments may be configured to perform, based on the first preference, a function related to the multiple first media items.

[0194] The processor according to one or more embodiments may be configured to, as at least a part of the function related to the multiple first media items, recommend, based on the first preference, at least one media item among the multiple first media items

[0195] The processor according to one or more embodiments may be configured to classify, based on the first preference, the multiple first media items.

[0196] The processor according to one or more embodiments may be configured to in case that a second AI model of other person is acquired from an external electronic device **202** or **203**, determine a second preference of each of the multiple first media items by using the second AI model. The processor according to one or more embodiments may be configured to perform, based on the second preference, a function related to at least one media item among the multiple first media items.

[0197] The processor according to one or more embodiments may be configured to in case that a third AI model of a specific person is acquired from a server **208**, determine a third preference of at least one media item among the multiple first media items by using the third AI model. The processor according to one or more embodiments may be configured to compare the first preference with the third preference to provide a guide including at least one of whether the at least one media item is preferred, a degree of preference, or correction information.

[0198] The processor according to one or more embodiments may be configured to identify whether the electronic device is in an idle state. The processor according to one or more embodiments may be configured to, based on the electronic device being in the idle state, train, based on the score and the feature, the first AI model.

[0199] The processor according to one or more embodiments may be configured to determine, based on the score, the specified media items among all media items stored in the memory as a media set for training of the first AI model.

[0200] The processor according to one or more embodiments may be configured to acquire a

weight related to the first AI model as a result of the training of the first AI model. The processor according to one or more embodiments may be configured to apply the weight to the main AI model to acquire the first AI model.

[0201] The processor according to one or more embodiments may be configured to identify an aesthetic score indicating the characteristic of each of the specified media items by using the main AI model. The processor according to one or more embodiments may be configured to train the first AI model by further using the aesthetic score in addition to the score and the feature.

[0202] The processor according to one or more embodiments may be configured to in case that a group is formed with multiple external electronic devices through the communication circuit, acquire an AI model from each of the multiple external electronic devices. The processor according to one or more embodiments may be configured to identify preferences of multiple media items related to the group by using the AI model. The processor according to one or more embodiments may be configured to transmit, based on the preference, at least one media item among the multiple media items to the multiple external electronic devices.

[0203] The processor according to one or more embodiments may be configured to acquire information on at least one media item from an external electronic device or a server. The processor according to one or more embodiments may be configured to train the first AI model by using information on the at least one media item.

[0204] The processor according to one or more embodiments may be configured to identify an attribute of each of the multiple first media items by using the first AI model. The processor according to one or more embodiments may be configured to classify, based on the attribute, the multiple first media items.

[0205] An operation method of an electronic device **201** according to one or more embodiments may include identifying a use pattern of specified media items among all media items stored in the electronic device, and determining, based on the use pattern, a score of each of the specified media items. The operation method of the electronic device according to one or more embodiments may include extracting a feature corresponding to a characteristic of each of the specified media items by using a main AI model stored in the electronic device. The operation method of the electronic device according to one or more embodiments may include acquiring a personalized first AI model trained based on the score and the feature. The operation method of the electronic device according to one or more embodiments may include determining, based on the first AI model, a first preference of each of multiple first media items. The operation method of the electronic device according to one or more embodiments may include performing, based on the first preference, a function related to the multiple first media items.

[0206] The performing of the function related to the multiple first media items according to one or more embodiments may include recommending, based on the first reference, at least one media item among the multiple first media items.

[0207] The performing of the function related to the multiple first media items according to one or more embodiments may include classifying, based on the first preference, the multiple first media items

[0208] The operation method of the electronic device according to one or more embodiments may include in case that a second AI model of other person is acquired from an external electronic device **202** or **203**, determining a second preference of each of the multiple first media items by using the second AI model. The operation method of the electronic device according to one or more embodiments may further include performing, based on the second preference, a function related to at least one media item among the multiple first media items.

[0209] The operation method of the electronic device according to one or more embodiments may further include in case that a third AI model of a specific person is acquired from a server **208**, determining a third preference of at least one media item among the multiple first media items by using the third AI model. The operation method of the electronic device according to one or more

embodiments may include comparing the first preference with the third preference to provide a guide including at least one of whether the at least one media item is preferred, a degree of preference, or correction information.

[0210] The operation method of the electronic device according to one or more embodiments may further include identifying whether the electronic device is in an idle state. The operation method of the electronic device according to one or more embodiments may further include based on the electronic device being in the idle state, training based on the score and the feature, the first AI model.

[0211] The operation method of the electronic device according to one or more embodiments may further include determining, based on the score, the specified media items among all media items stored in the electronic device as a media set for training of the first AI model.

[0212] A non-transitory recording medium **130** according to one or more embodiments may store instructions which can perform identifying a use pattern of specified media items among all media items stored in an electronic device **201**, and determining, based on the use pattern, a score of each of the specified media items, extracting a feature corresponding to a characteristic of each of the specified media items by using a main AI model stored in the memory, acquiring a personalized first AI model trained based on the score and the feature, determining, based on the first AI model, a first preference of each of multiple first media items, and performing, based on the first preference, a function related to the multiple first media items.

[0213] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to one or more embodiments of the disclosure, the electronic devices are not limited to those described above.

[0214] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0215] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to one or more embodiments, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0216] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may

invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0217] According to one or more embodiments, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0218] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. An electronic device comprising: memory storing one or more instructions; a communication circuit; and a processor operatively coupled to the memory and the communication circuit, wherein the one or more instructions, when executed by the processor, cause the electronic device to: identify a use pattern of specified media items from a plurality of media items stored in the memory, determine, based on the use pattern, a score of each of the specified media items, extract, using a main AI model stored in the memory, a feature corresponding to a characteristic of each of the specified media items, acquire a first AI model trained based on the score and the feature, based on the first AI model, determine a first preference of each of first media items from the plurality of media items, and based on the first preference, perform a function related to the first media items.
2. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to, as at least a part of the function related to the first media items, recommend at least one media item the first media items based on the first preference.
3. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to, based on the first preference, classify the first media items.
4. The electronic device of claim 1, wherein the processor is configured to: based on determining a second AI model of another person is acquired from an external electronic device, determine, using

the second AI model, a second preference of each of the first media items; and based on the second preference, perform a function related to at least one media item among the first media items.

5. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to: based on determining a third AI model of a specific person is acquired from a server, determine, using the third AI model, a third preference of at least one media item from the first media items, and compare the first preference with the third preference to provide information comprising at least one of whether the at least one media item is preferred, a degree of preference, or correction information.

6. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to: identify whether the electronic device is in an idle state, and based on identifying that the electronic device is in the idle state, train the first AI model based on the score and the feature.

7. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to, based on the score, determine the specified media items from the plurality of media items stored in the memory as a media set for training the first AI model.

8. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to: acquire, based on the training of the first AI model, a weight related to the first AI model, and apply the weight to the main AI model to acquire the first AI model.

9. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to: Identify, using the main AI model, an aesthetic score indicating the characteristic of each of the specified media items, and train the first AI model using the aesthetic score in addition to the score and the feature.

10. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to: based on determining a group is formed with a plurality of external electronic devices through the communication circuit, acquire an AI model from each of the plurality of external electronic devices, identify, using the AI model, one or more preferences of multiple media items related to the group, and based on the one or more preferences, transmit at least one media item from the multiple media items related to the group to the plurality of external electronic devices.

11. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to: acquire information on at least one media item from an external electronic device or a server, and train, using the information on the at least one media item, the first AI model.

12. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to: identify, using the first AI model, an attribute of each of the first media items, and based on the attribute, classify the first media items.

13. The electronic device of claim 1, wherein the one or more instructions, when executed by the processor, cause the electronic device to identify the use pattern based on at least one of viewing, sharing, editing, deleting, a favorites setting, or a background setting of the specified media items.

14. An operation method of an electronic device, the method comprising: identifying a use pattern of specified media items from a plurality of media items stored in the electronic device; determining, based on the use pattern, a score of each of the specified media items; extracting, using a main AI model stored in the memory, a feature corresponding to a characteristic of each of the specified media items; acquiring a first AI model trained based on the score and the feature; based on the first AI model, determining a first preference of each of first media items from the plurality of media items; and based on the first preference, performing a function related to the first media items.

15. The method of claim 14, wherein the performing of the function related to the first media items

further comprises recommending, based on the first preference, at least one media item from the first media items.

16. The method of claim 14, further comprising: based on the first preference, classifying the first media items.

17. The method of claim 14, further comprising: based on determining a second AI model of another person is acquired from an external electronic device, determining, using the second AI model, a second preference of each of the first media items; and based on the second preference, performing a function related to at least one media item among the first media items.

18. The method according to claim 14, further comprising: based on determining a third AI model of a specific person is acquired from a server, determining, using the third AI model, a third preference of at least one media item from the first media items; and comparing the first preference with the third preference to provide information comprising at least one of whether the at least one media item is preferred, a degree of preference, or correction information.

19. The method according to claim 14, further comprising: identifying whether the electronic device is in an idle state; and based on identifying that the electronic device is in the idle state, training the first AI model based on the score and the feature.

20. A non-transitory computer readable medium having instructions stored therein, which when executed by a processor in an electronic device, cause the processor to execute a method comprising: identifying a use pattern of specified media items from a plurality of media items stored in the electronic device; determining, based on the use pattern, a score of each of the specified media items; extracting, using a main AI model stored in the memory, a feature corresponding to a characteristic of each of the specified media items; acquiring a first AI model trained based on the score and the feature; based on the first AI model, determining a first preference of each of first media items from the plurality of media items; and based on the first preference, performing a function related to the first media items.
